Lux Zhang

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Experience

AEVEX Maritime Systems *(formerly BlackSea Technologies)* — Baltimore, MD, USA

Software Engineer Team Lead — *Jan 2026 – Present*

- Led 6 engineers shipping embedded software to 300+ fielded vessels, driving roadmap, sprints, and design review.
- Shipped releases to 6 vessels in live overseas operations spanning autopilot, navigation, and autonomy integration.
- Architected a hydrofoil prototype in 8 weeks from concept to on-water trials on an ArduPilot-based stack.
- Built single-operator control of 5 devices over 4 protocols via a safety-interlocked C++ daemon and TypeScript UI.
- Integrated 2 alternative pose and timing sensors into navigation to improve GPS-denied operation.
- Delivered control of 10 independent payload release tubes via a full-stack app (FastAPI, React, TypeScript).
- Authored 3 interface and requirements specs for third-party autonomy integrators and internal payload teams.
- Increased team output by ~50% by championing Claude Code adoption with 1:1 workflow coaching.
- Cut later-round false positives by ~40% by designing and running technical interviews for software hires.

Software Engineer — *Jan 2025 – Jan 2026*

- Migrated fleet-wide vessel comms from plain text to Protobuf, shrinking messages and ending schema-change bugs.
- Engineered a 50 Hz multithreaded C++ gateway linking vessel control to third-party AI navigation via DDS/NNG.
- Designed the J1939/NMEA 2000 CAN integration across engines, navigation, power distribution, and sensors.
- Decoupled development from hardware delivery with Dockerized CAN simulators that caught bugs pre-trial.
- Slashed vessel provisioning time by ~75% with Python and Bash tooling for setup and on-vessel debugging.
- Owned E-stop handling and control authority arbitration around a safety controller for multi-vessel operation.

Junior Software Engineer — *Jan 2023 – Jan 2025*

- Ported a 50,000-line embedded C codebase between two RTOS-based ECUs with zero vessel control downtime.
- Implemented engine control and ignition sequencing from scratch for 2 outboard families on 2 new platforms.
- Analyzed vessel telemetry with Python, pandas, and Matplotlib in Jupyter to diagnose system behavior.
- Spearheaded the move of internal comms to DDS in modern C++, introducing memory-safe practices.
- Halved build times by replacing GNU Automake with modular CMake and moving dependencies to FetchContent.

Federal Motor Carrier Safety Administration — Remote

Data Analyst Intern — *May 2020 – Aug. 2020*

- Identified highest-risk road segments for large-vehicle traffic across 23 counties using ArcGIS and Python.
- Automated statewide safety reports from crash data ETL pipelines, delivered to the chief safety officer.

Laureate Education — Columbia, MD, USA

Data Science Intern — *May 2019 – Aug. 2019*

- Consolidated 8 years of enrollment data from Oracle into SQL Server with Python and SQL ETL pipelines.
- Taught an internal seminar on Python data analysis and ML to onboard 20+ R/SAS analysts.

Education

Cornell University — Ithaca, NY

Bachelor of Science in Computer Science, cum laude; GPA: 3.6 — *Aug. 2018 – May 2022*

- Coursework: Data Structures and Algorithms, Operating Systems, Computer Networks, Digital Logic
- Minor in English

Skills

Languages: C++, C, Python, TypeScript, SQL
Robotics & Embedded: DDS, RTOS, CAN/SocketCAN, J1939, NMEA 2000, Protobuf, ArduPilot, QGroundControl
Tools: Linux, Bash, Git, Docker, CMake, GitLab, JIRA, Claude Code, OpenAI Codex
Web: FastAPI, React, Tailwind CSS, Node.js, Vite, PostgreSQL, SQLite